



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SONASUN TRADERS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

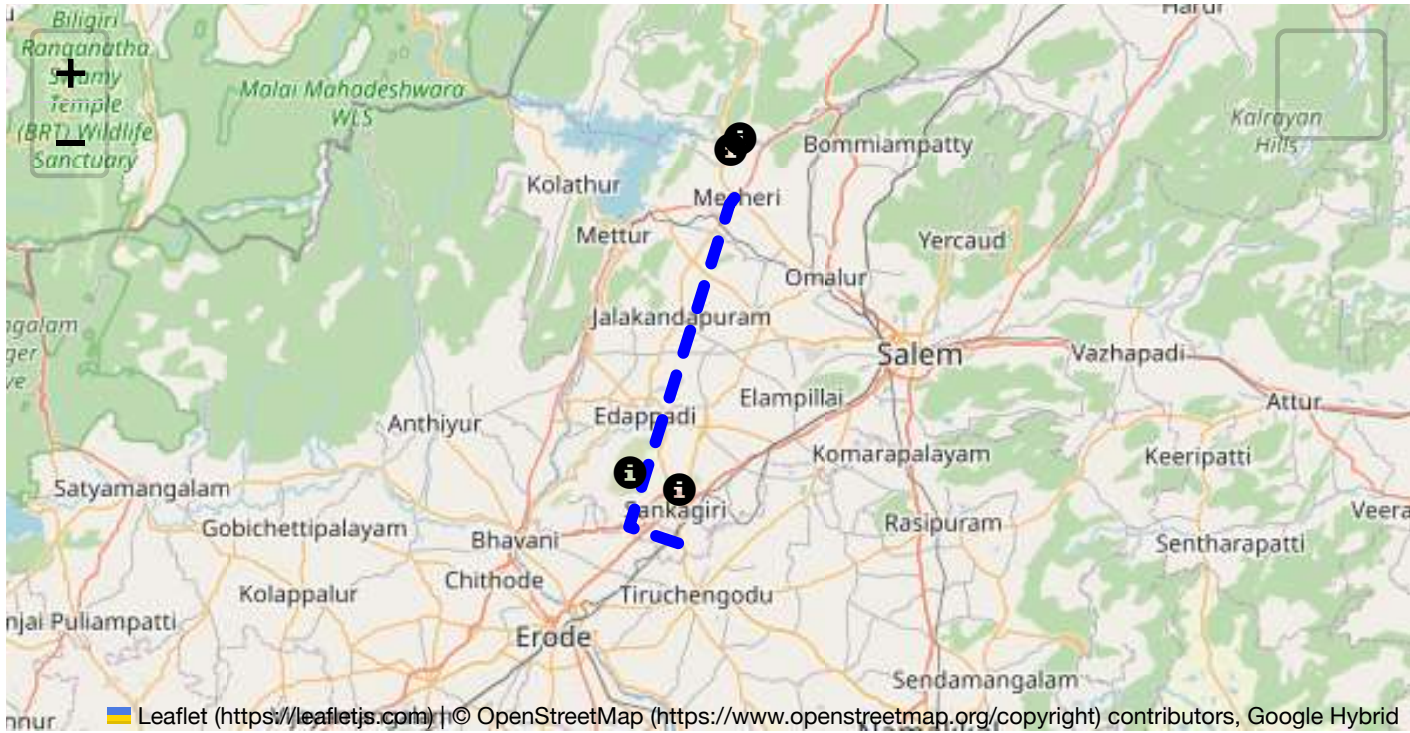
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

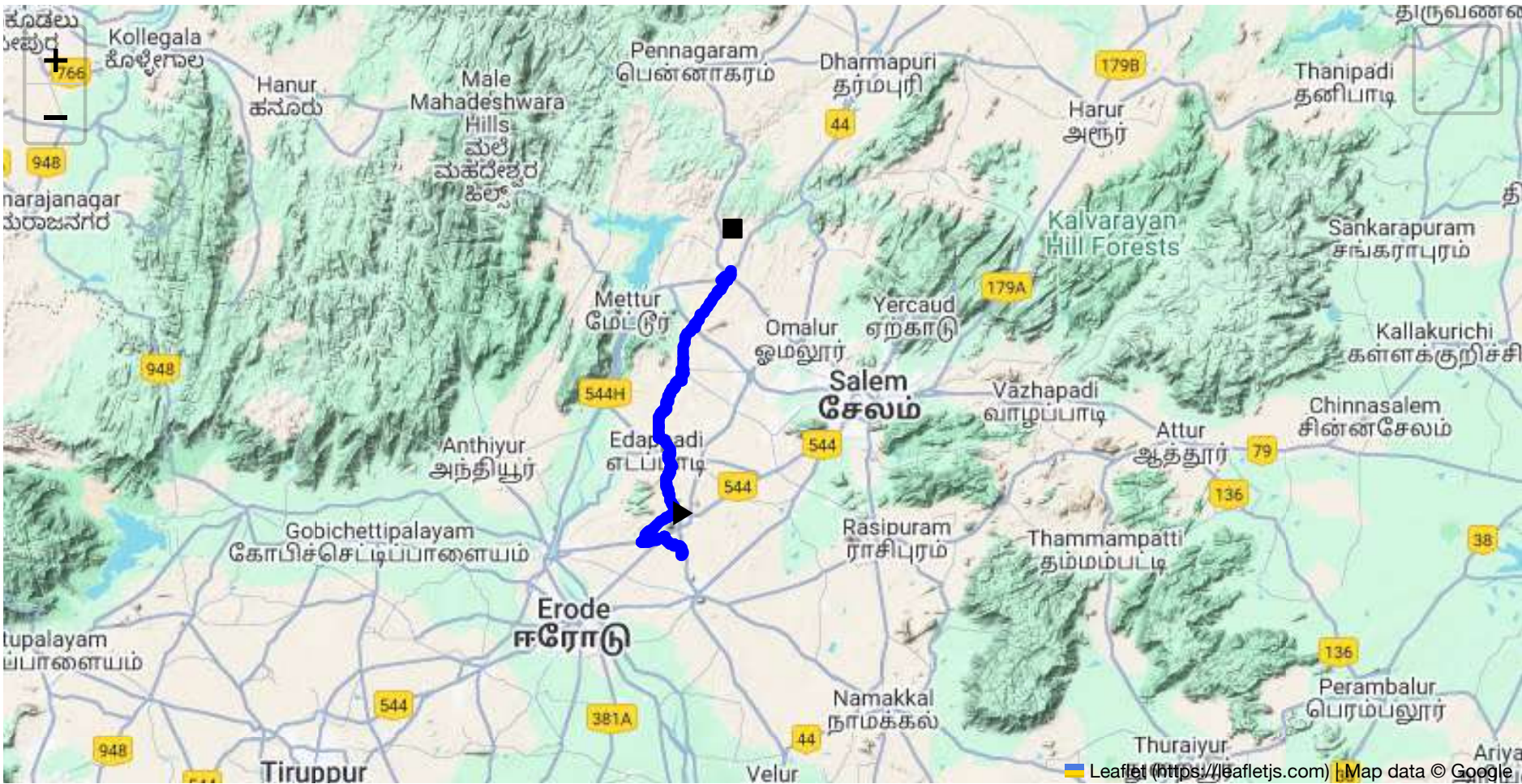
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 62.76 km
Estimated Duration: 1.6 hours
Adjusted Duration (Heavy Vehicle): 1.9 hours

Start: (11.4381, 77.8734)
End: (11.840141, 77.946205)



Welcome to the Journey Risk Management Study

Route Overview

The route spans approximately 62.76 kilometers, beginning in Sangagiri and ending in Mecheri, passing through Padaiveedu and Mettur. This journey involves traversing regional roads, including Bhavani Main Road and State Highway 158 (SH 158).

Weather Conditions

Typical weather includes hot and humid conditions, especially from March to June, with potential for monsoon rains between June and September. This may lead to water logging and slippery road conditions, particularly in low-lying areas.

Traffic Patterns

Traffic congestion is likely during morning (8 AM - 10 AM) and evening peak hours (5 PM - 7 PM), especially in populated areas like Sangagiri and Mettur. Road interchanges and market areas in these towns could experience heavy traffic, making navigation challenging for heavy vehicles.

Road Quality and Infrastructure

The roads are generally paved, but there may be sections with potholes or uneven surfaces, especially near rural villages. Road signage might be insufficient in places, requiring careful navigation by drivers.

Alternative Routes for Emergencies

In case of emergencies, an alternative route could be via the Erode-Chithode-Pallipalayam Road connecting to NH544, though this might involve longer travel time and detours depending on the starting and ending points within villages.

Local Regulations on Hazardous Material Transport

Tamil Nadu imposes strict regulations on the transport of hazardous materials, requiring permits and adherence to safety protocols. Restrictions may include travel time limits and specific routes to be used, often aligning with national guidelines.

Historical Incidents

Instances of accidents involving heavy vehicles often occur due to road quality and weather conditions. Mettur, being an industrial zone, may have stricter checks and potential for incidents of hazardous material spillage or other mishaps.

Environmental Considerations

The route passes through areas adjacent to water bodies, necessitating careful handling of hazardous materials to prevent contamination. Proximity to the Mettur Dam requires extra caution.

Communication Coverage

Telecommunication coverage might be inconsistent, particularly in the more rural or hilly parts of the route. Notable dead zones may exist around the more remote or densely forested areas.

Emergency Response Times

Response times are generally longer in rural sections, ranging from 30 to 60 minutes, but these can be reduced to approximately 15-20 minutes near towns like Sangagiri and Mettur due to better infrastructure.

Overall Risk Assessment

- High-risk areas:** Road quality and weather conditions in rural and low-lying zones.
- Moderate-risk areas:** Congestion in towns and regulatory obstacles for hazardous materials.
- Low-risk areas:** Most of the road sections between towns where infrastructure support is better.

The journey requires strategic planning around peak traffic times, adherence to local laws, awareness of environmental concerns, and preparation for variable weather conditions to ensure safety and compliance.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.58300, 77.85813	15 KM/Hr
1	Roundabout	High	11.60348, 77.83967	15 KM/Hr
2	Turn	Medium	11.43822, 77.87348	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44029, 77.87544	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
5	Turn	Medium	11.44900, 77.87406	30 KM/Hr
6	Turn	Medium	11.45358, 77.85781	30 KM/Hr
7	Turn	High	11.45353, 77.85697	15 KM/Hr
8	Turn	Medium	11.45853, 77.85139	30 KM/Hr
9	Turn	Medium	11.45906, 77.85132	30 KM/Hr
10	Turn	Medium	11.46307, 77.84941	30 KM/Hr
11	Turn	Medium	11.46316, 77.84921	30 KM/Hr
12	Turn	High	11.45542, 77.81725	15 KM/Hr
13	Turn	High	11.45636, 77.81678	15 KM/Hr
14	Blind Spot	Blind Spot	11.49397, 77.86740	10 KM/Hr
15	Turn	Medium	11.49971, 77.86468	30 KM/Hr
16	Turn	Medium	11.50885, 77.86037	30 KM/Hr
17	Turn	Medium	11.52571, 77.85522	30 KM/Hr
18	Turn	Medium	11.52744, 77.85582	30 KM/Hr
19	Turn	Medium	11.55049, 77.85513	30 KM/Hr
20	Turn	Medium	11.55460, 77.85624	30 KM/Hr
21	Turn	High	11.57893, 77.85430	15 KM/Hr
22	Turn	Medium	11.57903, 77.85441	30 KM/Hr
23	Turn	Medium	11.60978, 77.84165	30 KM/Hr
24	Turn	Medium	11.61795, 77.84020	30 KM/Hr
25	Turn	Medium	11.63085, 77.84072	30 KM/Hr
26	Turn	Medium	11.64055, 77.84600	30 KM/Hr
27	Turn	Medium	11.64173, 77.84878	30 KM/Hr
28	Turn	Medium	11.64547, 77.85129	30 KM/Hr
29	Turn	Medium	11.65486, 77.85335	30 KM/Hr
30	Turn	Medium	11.67310, 77.86031	30 KM/Hr
31	Turn	Medium	11.68196, 77.86690	30 KM/Hr
32	Turn	High	11.68492, 77.86898	15 KM/Hr
33	Turn	Medium	11.68463, 77.87262	30 KM/Hr
34	Turn	Medium	11.68486, 77.87288	30 KM/Hr
35	Turn	High	11.69469, 77.87589	15 KM/Hr
36	Turn	High	11.69558, 77.87439	15 KM/Hr
37	Turn	Medium	11.69908, 77.87472	30 KM/Hr
38	Turn	Medium	11.69918, 77.87487	30 KM/Hr
39	Turn	High	11.76019, 77.89047	15 KM/Hr
40	Turn	Medium	11.75964, 77.89200	30 KM/Hr
41	Blind Spot	Blind Spot	11.75947, 77.89214	10 KM/Hr
42	Blind Spot	Blind Spot	11.82791, 77.94044	10 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
43	Turn	Medium	11.82810, 77.94014	30 KM/Hr
44	Blind Spot	Blind Spot	11.82769, 77.93262	10 KM/Hr
45	Turn	Medium	11.83363, 77.94443	30 KM/Hr
46	Turn	High	11.83386, 77.94485	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Parusuraman Hospital	11.697164, 77.877792	30 km/h	Medium
2	hospital	JP Nature Cure Hospital Siddha and Naturopathy	11.7002949, 77.8770897	30 km/h	Medium
3	hospital	SSS Hospital	11.7501471, 77.8832809	30 km/h	Medium
4	clinic	DSR Clinic	11.7603176, 77.8908853	30 km/h	Medium
5	hospital	Srilakshmi Hospital	11.7601, 77.892272	30 km/h	Medium
6	hospital	Government Hospital	11.758158, 77.894528	30 km/h	Medium
8	hospital	Vasu Speciality Hospital	11.8331097, 77.943169	30 km/h	Medium
9	hospital	Rajamanickam Hospital	11.8332408, 77.9431081	30 km/h	Medium
10	hospital	Child Hospital	11.833001, 77.943415	30 km/h	Medium
11	hospital	Naveen Nursing Home	11.8335355, 77.9437441	30 km/h	Medium
12	hospital	Sri Sakthi Hospital	11.835714, 77.9451014	30 km/h	Medium
13	clinic	Sri Krishna Clinic	11.835443, 77.9450587	30 km/h	Medium
14	hospital	Selvakumar Hospital	11.836215, 77.945842	30 km/h	Medium
15	clinic	SK Clinic	11.83619, 77.9457752	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
7	college	Kavery Educational Institutions	11.8255319, 77.9342256	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.58300, 77.85813



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.60348, 77.83967



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45358, 77.85781



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45853, 77.85139



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45906, 77.85132



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46316, 77.84921



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45636, 77.81678



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49397, 77.86740



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49971, 77.86468



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50885, 77.86037



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52571, 77.85522



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52744, 77.85582



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55049, 77.85513



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55460, 77.85624



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.57893, 77.85430



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57903, 77.85441



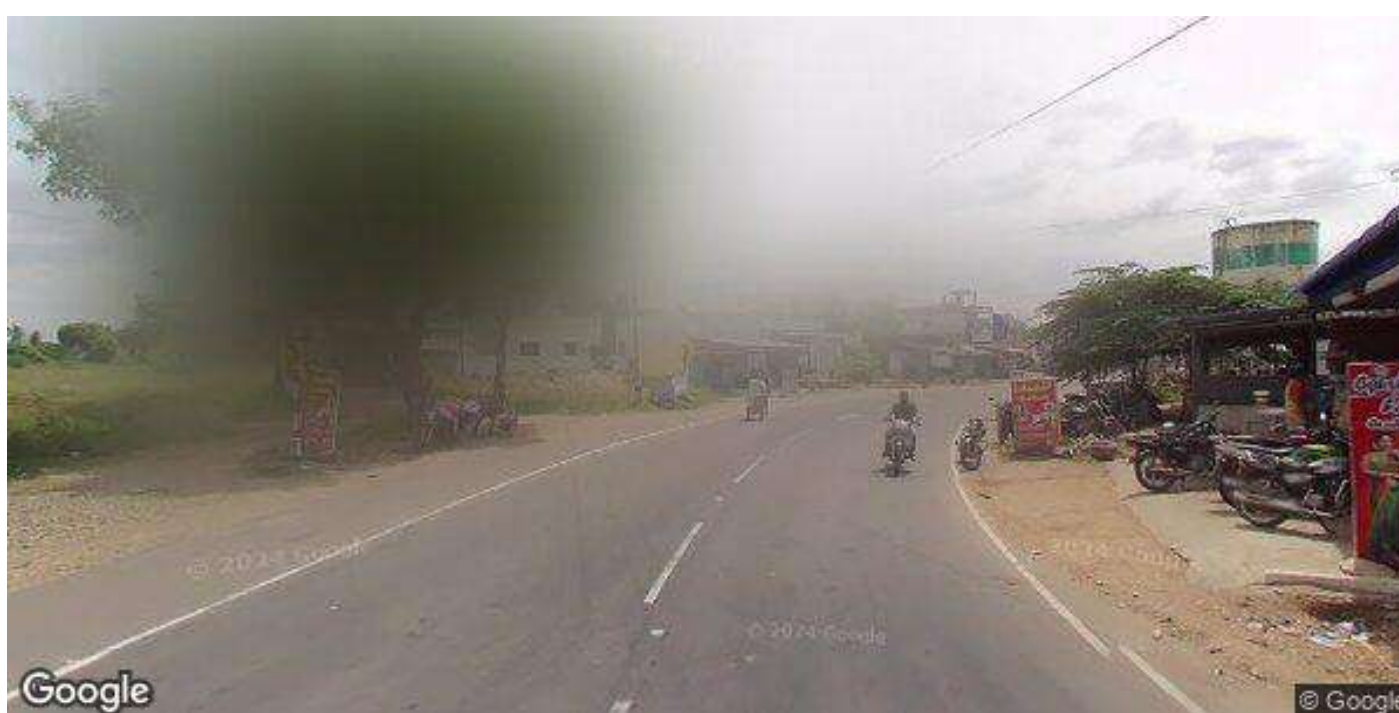
Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60978, 77.84165



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61795, 77.84020



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63085, 77.84072



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64055, 77.84600



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64173, 77.84878



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64547, 77.85129



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65486, 77.85335



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67310, 77.86031



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68196, 77.86690



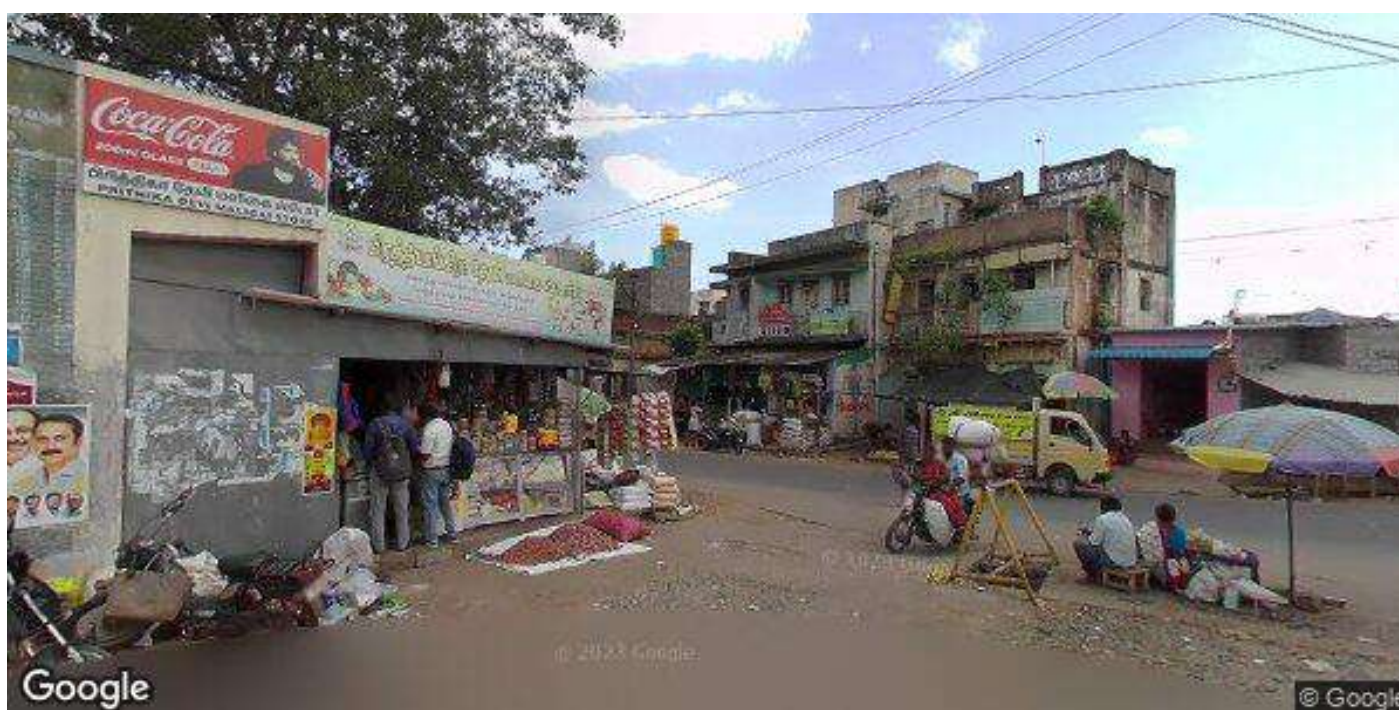
Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.68492, 77.86898



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68463, 77.87262



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68486, 77.87288



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.69469, 77.87589



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.69558, 77.87439



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69908, 77.87472



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69918, 77.87487



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.76019, 77.89047



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.75964, 77.89200



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.75947, 77.89214



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.82791, 77.94044



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.82810, 77.94014



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.82769, 77.93262



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.83363, 77.94443



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.83386, 77.94485
